

SUMMARY

Automation & AI Engineer | Python Developer | Machine Learning Specialist | Computer Vision

Results-driven engineer with 4+ years of experience in automation, Python development, and end-to-end deployment of machine learning models. Expert in computer vision, data science, and process automation for real-time, high-impact solutions. Proven track record of reducing manual effort by 70–95% and optimizing workflows using EDI, cloud platforms (AWS, GCP), and advanced ML frameworks. Certified in Machine Learning (Stanford), Deep Learning, and Data Science.

WORK EXPERIENCES

Automation and AI Engineer, Crystals (Columbus, GA and St. Louis, MO)

Mar 2024 – Apr 2025

- Automated EDI workflows reducing manual processing time by 70% and ensuring secure, real-time electronic delivery of transactional documents.
- Designed and deployed autonomous loadout solutions, cutting project costs by 55% and manual effort by 95% through in-house technical innovation.
- Implemented truck weighing software, reducing manual steps by 20% and minimizing human error, while automating ticket information flow through custom email automation scripts, resulting in significant productivity gains.
- Integrated Siemens historian and HPLC data into live dashboards, empowering data-driven decision-making and boosting production capacity.
- Led IT infrastructure modernization, implementing Palo Alto Networks for secure remote access and integrating AI-powered productivity tools across Google Workspace and Microsoft 365.
- Developed and launched a multi-level purchase requisition and financial tracking system, improving supply chain efficiency and spend visibility.
- Created 5+ dashboards for real-time monitoring, improving data accessibility, and decision-making speed.

Data Analyst, Realsoft (Herndon, VA – Freelance)

Jan 2024 – Mar 2024

- Analyzed large-scale EHR datasets using SAS, applying advanced statistical models to improve patient outcome predictions by 15%.
- Built interactive PowerBI dashboards for healthcare KPIs, reducing patient wait times by 20% and optimizing resource allocation.

Graduate Research Assistant (Computer Vision), CSU (Columbus, GA – Hybrid)

Sep 2022 – Dec 2023

- Developed a real-time video surveillance system with object detection and tracking, reducing manual monitoring by 80% and enabling automated alerts.
- Built automated data annotation pipelines for imagery datasets, streamlining data preparation and cleaning.
- Trained and optimized computer vision models on NVIDIA Jetson, achieving 0.84 mAP and 15% improved GPU efficiency using CUDA.
- Managed model deployment in production, optimizing GPU computing efficiency by 15% through CUDA and implementing deep learning techniques for low power usage.
- Engineered Python APIs for real-time data transfer to Oracle databases, enhancing system integration and response time.

Application Development Associate, Accenture (Maharashtra, India – Remote)

Nov 2020 – Jun 2022

- Automated data pipelines using Hadoop, AWS Athena, Docker, and Jenkins, enabling efficient CI/CD for financial analytics.
- Managed AWS-based data lakes, performing advanced analytics with APIs and Airflow ETL tools.
- Developed a custom code validation tool, aligning with client requirements and delivering a production-ready executable to improve code quality.
- Trained NLP models to extract requirements from compliance documents, improving project prioritization and decision-making.
- Collaborated on financial data research, developing a comprehensive data visualization dashboard for reporting.
- Developed RPA solutions for web/database testing, accelerating QA cycles, and saving 100+ hours of manual effort weekly for each user.

SKILLS

Core Competencies

- Machine Learning, Python Programming, Computer Vision, Data Science, Automation Engineering, Cloud Computing (AWS, GCP), DevOps (Docker, Jenkins, Git), EDI Solutions, Real-time Systems, Dashboard Development, API Integration, RPA, NLP, Statistical Analysis, Data Visualization (PowerBI, Tableau), Linux Administration, Cross-functional Collaboration, Project Leadership

Technical Skills

- **Programming:** Python, R, SQL, Java, JavaScript
- **ML Frameworks:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy
- **Computer Vision:** OpenCV, YOLO, DeepSORT
- **Generative AI:** Langchain, RAG, BERT, CrewAI, BeamAI
- **DevOps:** Docker, Jenkins, Git, Airflow
- **Cloud:** AWS, GCP
- **Data Visualization:** PowerBI, Tableau, Grafana
- **Environments:** Firebase Studio, Cursor, VS Code, RStudio, Linux
- **Productivity:** Jira, Confluence, Monday
- **Administration:** Google Workspace, Microsoft 365

Soft Skills

- Process Improvement, Problem-solving, Team Leadership, Documentation Management, Escalation Management, Communication

EDUCATION

Columbus State University (Georgia, USA)

Aug 2022 - Dec 2023

Master of Science in Computer Science (Artificial Intelligence and Machine Learning)

GPA 4.00 / 4.00

Gujarat Technological University - LDCE (Gujarat, India)

Jul 2016 - Jul 2020

Bachelor of Engineering in Electronics and Communication

GPA 8.38 / 10.00

CERTIFICATIONS

Machine Learning (Stanford University), **R-Programming** (John Hopkins University), **Deep Learning** Specialization (Stanford University), **Data Science** (HarvardX), **Big Data** with SQL (Cloudera), **AWS** Fundamentals (Coursera), **GCP**, **Google AI Essentials** (Coursera).

PROJECTS

Gen AI using Large Language Models (Georgia, USA)

Open Source Project - Mar 2024

- Fine-tuned and deployed LLMs (BERT, RAG) in production using Docker and cloud platforms, increasing domain-specific NLP accuracy and scalability.
- Researched and implemented transformer-based architectures to enhance contextual understanding and relevance in real-world NLP applications.

Multi-Object Tracker Surveillance System (Georgia, USA)

Open Source Project - Dec 2023

- Developed and implemented a DeepSORT-based multi-object tracking system, improving real-time surveillance precision by 6% and optimizing GPU deployment on NVIDIA Jetson by 15%.

Self-Driving Raspberry Pi Car - Computer Vision and IoT (Georgia, USA)

Academic Project - Dec 2022

- Engineered an autonomous navigation system on Raspberry Pi with real-time object detection and path planning using machine learning algorithms.
- Integrated Tensor Processing Units (TPU) to accelerate inference, reducing system latency by 20%.
- Designed a Linux-based control system for seamless hardware-ML interaction.

Visualizing Citi Bike Trips with Tableau (Gujarat, India)

Coursera Certified Project - Oct 2020

- Designed and published interactive Tableau dashboards to visualize and analyze urban mobility data.

Bird-Strike Prevention System - Computer Vision and IoT (Gujarat, India)

Academic Project - Jun 2018

- Created a real-time bird detection and prevention system prototype using Caffe2, OpenCV, and Arduino for aviation safety.